Ruban T(2,3) Reproduction Falsifier

Numerical pilot | SMOKE | v0.1.0 | 20261030T071132181755Z

SCIENTIFIC STATUS: DEVELOPMENT ONLY - PUBLICATION REPRODUCTION UNRESOLVED

1. Claim and experiment

Attempt to reproduce Ruban (2018), Long-lived quantum vortex knots, arXiv:1801.04822, equations (6)-(9), for a trefoil T(2,3). The paper presents a long-lived configuration at B0=0.154, Lambda=1.4+0.5 ln(200), and delta=0.05, with survival beyond t=300 and axial travel above 1200 R0. This package is a filament-only model and makes no full 3D Euler or physical stability claim.

2. Implemented operator

Normalized R0=1, Gamma=2*pi. Velocity = 0.5 * integral [X_beta(source) cross (X(target)-X(source))] / [(distance)^2+delta^2]^(3/2) d(beta_source) + 0.5*lambda_tilde*kappa*binormal, with lambda_tilde=Lambda+ln(delta). Fourier spectral beta derivatives, periodic trapezoid rule, RK4 time integration. No external forcing, remeshing, smoothing or artificial damping. Delta is a numerical regularization and is distinct from physical core size xi.

Parameter	Actual
B0 scan	0.154
Lambda	4.049158683
delta/R0	0.0500
N, dt	64, 0.001
target t	0.025
initial perturbation	1e-05

3. Measured outputs

B0	t attained	Energy drift	Axial travel	Stop reason
0.154	0.025	2.07e-06	0.0889	TIME_LIMIT

4. Gate ledger

G0_PROVENANCE: PASS

G1_RING_SYMMETRY: PASS

G2_NUMERIC_ENERGY: PASS

G3_RUBAN_T300_AND_TRAVEL1200: UNRESOLVED

G3_PUBLICATION_ENERGY_7DIGITS: UNRESOLVED

G4_STABILITY_WINDOW: UNRESOLVED

G5_CONVERGENCE: UNRESOLVED

G6_PAPER_REPRODUCTION: UNRESOLVED

G7_EULER_FINITE_CORE: NOT_TESTED

5. Interpretation and stop criteria

A PASS for solver symmetry or short-time energy checks is not a PASS for Ruban replication. Only the canonical B0=0.154 case may satisfy the paper-specific lifetime gate. An operational deformation threshold of 0.25 R0, relative energy drift cap of 0.001, and segment-length ratio cap of 10 are new explicit numerical stop criteria, NOT calibrated to the author's unpublished lifetime algorithm. Publication-level validation requires independent spatial/time convergence and full-duration integration.

6. Scope exclusions

No 3D finite-core Euler field, internal core twist, background parallel filaments, exact equilibrium/Floquet analysis, reconnection algorithm or physical clock mechanism is modeled. Filament topology is assumed for smooth evolution only while no collisions/singularities occur. Fixed-grid parameter-space window comparison remains unresolved.

7. Provenance and citation

V. P. Ruban, Long-Lived Quantum Vortex Knots, JETP Letters 107 (2018), DOI: 10.1134/S0021364018050120; arXiv:1801.04822v2. Extended geometry: Ruban, Quasi-Stable Configurations of Torus Vortex Knots and Links, JETP 127 (2018), DOI: 10.1134/S106377611809008X. E010-v0.4.0 combined atlas used as geometry-only context, not a computed dynamics provider.

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